

KAZI SAMIN YASAR ALAM

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PhD Applicant, Fall 2027

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RESEARCH STATEMENT

I am interested in data-centric NLP and in the data systems behind it: how data is collected, validated, synchronized and served, and how data quality shapes the evaluation of language models. My work spans building the MixSarc annotated code-mixed corpus, evaluating LLMs in a zero-shot dialogue study, and contributing to offline-first data collection and national dashboard platforms, which motivates my interest in distributed data systems, data management and data-centric NLP.

EDUCATION

B.Sc. in Computer Science & Engineering, Islamic University of Technology (IUT) Jun 2021 – Oct 2025
CGPA: **3.71/4.00**

Higher Secondary Certificate (Science), Notre Dame College, Dhaka Jul 2018 – Apr 2020
GPA: **5.00/5.00**

PUBLICATIONS

K. S. Y. Alam, M. T. Chowdhury, T. Ahmed, A. Abrar, and M. R. Haque. *MixSarc: A Bangla-English Code-Mixed Corpus for Implicit Meaning Identification*. **Preprint; under review at PLOS.** [\[Preprint\]](#)

RESEARCH EXPERIENCE

MixSarc: Bangla-English Code-Mixed Corpus for Implicit Meaning Identification | [Preprint](#) Undergraduate Thesis

- Built the first public Bangla-English code-mixed corpus annotated for humor, sarcasm, offensiveness and vulgarity: 9,087 sentences from 31,129 scraped social media posts, filtered with script-based rules and an mBERT code-mix classifier.
- Recruited and coordinated 18 annotators using a qualification test and written guidelines; labels come from triple annotation with majority voting (Fleiss' $\kappa = 0.66$).
- Benchmarked fine-tuned Banglish-BERT and Gemma-2B (QLoRA) and five zero-shot LLMs; humor detection reached F1 0.708, while sarcasm, offensiveness and vulgarity remained hard due to class imbalance.
- Applied a MixSarc-tuned classifier to BnSentMix and estimated that about 42% of negative-sentiment samples show sarcastic characteristics.

Zero-Shot Task-Oriented Dialogue (Collaborative Research Project), with Adib Mosharrof (then PhD student, University of Kentucky) Sep 2024 – Jan 2025

- Evaluated GPT-4o as a zero-shot baseline on three unseen Schema-Guided Dialogue (SGD) domains (Hotels, Music, Restaurants), measuring API-call invocation, API-call correctness and response quality (GLEU).
- Analyzed failure cases: the model invoked API calls reliably but produced fully correct calls less than half the time.
- Contributed to prompt design and baseline comparison.

STANDARDIZED TESTS

GRE: 320 total (Quantitative 165, Verbal 155, Analytical Writing 3.5)

IELTS: 8.0 overall (Listening 8.5, Reading 9.0, Writing 6.5, Speaking 7.5)

PROFESSIONAL EXPERIENCE

Business Analyst, Dynamic Solution Innovators Ltd.

Nov 2025 – Present

- Translated stakeholder needs from client meetings into technical specifications, user guides, and process flows for development teams.
- Maintained a centralized SQL database of resource skills and CVs for project staffing.

Data Operations Analyst, Commure Bangladesh

Sep 2025 – Nov 2025

- Imported, validated and reconciled large volumes of operational and financial data from multiple sources for the Commure RCM Ops Colossus US team's healthcare revenue cycle management processes.
- Improved data pipelines and reporting workflows, building automated dashboards that reduced manual reporting effort.

PROJECTS

EARN Information Management System (EIMS)

SQL, Data Migration, Data Validation, Offline-First Mobile Data Collection

- Contributed to the World Bank-financed EARN Project of the Government of Bangladesh, targeting the selection and training of 900,000 NEET youth; led requirements elicitation and end-to-end workflow design for the centralized MIS development with mobile survey modules featuring offline and geolocation capabilities.
- Migrated the government BBS geographic code database (about 36,000 hierarchical codes, division to village/union level) into the platform's schema using SQL; validated the data and found name mismatches, duplicates and missing entries in about 500 codes.
- Produced SRS and SDD documents and UML diagrams (use case, sequence, state) for all modules.

National Dashboard Framework (NDF), a2i

Apache Superset, Data Visualization, System Design Documentation

- Contributed to an Apache Superset-based dashboard platform that visualizes ministry performance trends and fluctuations including an AI feature that summarizes charts.
- Produced the SRS and use case diagrams for spatial visualization, time-series analysis and thematic data filtering.

TECHNICAL SKILLS

Programming Languages: Python, C++, Java, SQL

Machine Learning & NLP: PyTorch, TensorFlow / Keras, Hugging Face Transformers, Scikit-learn, Transformer-based Models, Word Embeddings, Model Fine-tuning, LLM API Integration

Data Tools & Scripting: Data Analysis, Data Validation, Web Scraping, Automation Scripting, Power BI, Retool, Facepager

Backend & Databases: MySQL, MongoDB, Firebase, Node.js

LEADERSHIP & ACTIVITIES

President, IUT Computer Society (IUTCS)

Jul 2024 – Sep 2025

- Led IUT's largest student-run computing organization (10+ executive teams, 100+ members); a year-long training pipeline resulted in 4 contests, 2 national hackathons, and a 15% rise in participation.

General Secretary, IUT Computer Society (IUTCS)

Jul 2023 – Jun 2024

- Managed operations for the IUT 11th National ICT Fest, the largest to date with 2,000+ participants, 12 events, and 19 sponsors.